

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS (NC State, May 2026). Built production **document processing pipelines** (PDF gen, email parsing, S3) and full-stack AI features at an AI startup -- TypeScript, Node.js, React, Python, Java, PostgreSQL, Redis, Docker, Kubernetes. Multi-model LLM agent tooling (Claude, GPT-4o, Gemini via MCP) and **full-stack ownership from DB schema to React UI**. Ready to ship document automation features at Instabase from day one.

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern

- Owned full backend of a production AI platform -- built **document processing pipelines** (PDF generation, email parsing via Microsoft Graph API, S3 file storage) and real-time event pipelines using **TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma ORM**, Auth0, Docker, and Sentry across 10+ shipped features.
- Engineered **multi-model AI agent tooling** across 3 providers and 8+ models (Anthropic Claude, OpenAI GPT-4o, Google Gemini) via MCP with Temporal.io durable workflow orchestration and structured LLM outputs -- provider-agnostic interfaces, 394 commits, PR reviews across the team.

Mphasis (HP Company) Jun -- Aug 2024

Software Developer Intern

- Built AI-driven claim automation workflow (OCR, NLP, ML) improving accuracy 25%; automated deployments on **AWS and GCP** with Grafana + Sentry observability and red-teaming support.

Capgemini Jun -- Aug 2023

AI/ML Intern

- Kafka + Spark + Hadoop pipelines for real-time data ingestion; cut processing time 15% and latency 20%.

PROJECTS

PulseRank -- Real-Time Leaderboard & Activity Feed 2026

Full-stack event-driven platform: **NestJS + Next.js + React**, MySQL outbox pattern, MongoDB TTL feed projections, Redis sorted sets for sub-ms leaderboard reads, Server-Sent Events for live updates. Multi-store consistency via Testcontainers; OpenTelemetry + Prometheus; GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, React, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

Inventory Reservation Service 2026

Transactional inventory API in **Java 21 + Spring Boot** preventing overselling via PostgreSQL row-level locking across multi-SKU reservations; idempotent endpoints, automated expiry job. Concurrency verified with Testcontainers; Flyway migrations, Spring Security, multi-stage Docker build, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker, GitHub Actions

AlgoPulse -- Agentic AI Platform 2025

Event-driven **LLM workflow orchestration** platform. Diagnosed API-polling bottleneck; pivoted to Supabase Edge Functions streaming AI reasoning directly to client. Temporal.io for durable workflow state + Kafka for fault-tolerant worker coordination -- 2,000 async inference req/sec without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

EDUCATION

MS Computer Science -- North Carolina State University Aug 2024 -- May 2026

AI, Operating Systems, Data Science, Algorithms, Software Engineering

B.Tech Computer Science -- Mumbai University Aug 2020 -- May 2024

CGPA 9.32/10 | Machine Learning, Big Data Infra, Data Warehousing, Algorithms

SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL, C

Stack: Node.js, React, Next.js, NestJS, Spring Boot, FastAPI, Express, Prisma ORM, Temporal.io, Kafka

AI/ML: OpenAI API, Claude API, Gemini API, LangChain, Hugging Face, PyTorch, RAG, MCP, LLM Evals

Infra: Docker, Kubernetes, AWS, GCP, PostgreSQL, Redis, MongoDB, Sentry, Prometheus, GitHub Actions

Awards: HackNC24 Top 10 (100+ teams), 6+ national hackathon wins, IEEE Published, Top 1% Sanjeevani (1100+ teams)